Ricardo Ribeiro

Senior Software Engineer · Full-Stack & Platform

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SUMMARY

Senior software engineer with 7+ years building and running production systems end to end: backend, frontend, cloud infrastructure and, increasingly, AI. I own platforms from the product feature down to CI/CD, Kubernetes, observability, databases and security. I have worked on police-intelligence platforms, subscription billing for the largest telecom in Argentina, multi-tenant HR software, internal financial systems and marketing automation. I have a B.Sc. in Computer Science and an MBA in Software Engineering (USP), and I'm currently doing a postgraduate degree in Applied AI Engineering.

EXPERIENCE

Group Wave Enterprise — Senior Software Engineer

Mar 2026 – Present · Contract · Remote

I own the product and the platform of a marketing-automation SaaS. I delivered 1,000+ merged pull requests across backend, frontend and infrastructure in about six months.

- **Performance:** Made application workflows more than 70% faster through refactoring, caching, BullMQ job deduplication and background workers for CPU-bound work.
- **Platform and Kubernetes:** Owned the cloud-native platform: Kubernetes (k3s), Kustomize, Ansible, PostgreSQL, Redis, RabbitMQ and MinIO. Workers run as dedicated deployments per queue and autoscale on queue depth with KEDA.
- **CI/CD:** Rebuilt CI/CD in three phases: GitHub Actions, then self-hosted ephemeral runners on Kubernetes, then a standalone Jenkins provisioned with Ansible and configured as code. CI is isolated from production workloads. PRs run only the affected tests, with diff-coverage gates. Each step is documented in ADRs. Profiling the pipeline cut backend PR time by 26% (lint and tests 380 s → 244 s), shrank the backend image by about 41% (2.5 GB → 1.5 GB) and cut one build step from about 285 s to 5 s.
- **Databases:** Led a PostgreSQL connection-management project (PgBouncer, connection budgets per worker type, idle-transaction cleanup, tuning). Added automated and encrypted off-site backups, and audited, least-privilege database access for engineers.
- **Observability and SRE:** Built the observability stack: OpenTelemetry tracing from the backend to the UI, Prometheus with Thanos, Loki, Grafana and Grafana Faro for real-user monitoring. Added SLO burn-rate alerting, runbooks, post-mortems and load tests.
- **Internal financial platform:** Led the design and built most of it (backend and frontend):
  - **Ledger and ERP:** multi-currency P&L at daily exchange rates, bank and card-account sync through queues with per-account progress, and automatic matching of internal transfers.
  - **Categorization:** a rules engine where the most specific rule wins and conflicts are flagged, plus bulk categorization.
  - **Finance operations:** commissions with monthly close snapshots, contractor payroll, invoices, tax forecasting, performance bonuses, accounting documents, a partner registry (CSV import, duplicate merge) and an activities suite (tasks, calendar, processes).
  - **Security:** designed with defense in depth: tenant isolation enforced by the database as well as the application, encrypted and masked sensitive data with audited access, a tamper-evident audit trail, step-up authentication for critical actions, time-bound access grants with access reports and security alerting, and automated security checks in CI.
- **Campaign automation:** Built a rules engine for ad campaigns with conditions, budget actions, scheduling, rule versioning and a full execution history. Also built a versioned external API with per-tenant keys and a published OpenAPI contract.
- **Incidents and audits:** Led root-cause analysis of production incidents (table bloat, disk I/O saturation, including a logging loop growing about 20 GB/h, and a runaway budget automation), turning each into fixes, guardrails and alerts. Led a multi-tenant readiness audit with a GO/NO-GO assessment and a phased 13-sprint plan.
- **Security:** Led platform hardening: secrets management, secret scanning and static analysis in CI, SSO for internal tools, and ingress hardening.
- **Architecture and AI:** Worked with an external architecture consultant on the move from microservices to a modular React/NestJS monolith. Built reusable LLM agents and skills and a shared Claude Code knowledge base for the team.

ErosAI (MV) — Software Consultant

Apr 2026 – Sep 2026 · Consulting · Remote

- **AI cost:** Cut monthly AI infrastructure costs by more than 90% by redesigning the AI architecture around a RAG pipeline on a vector database. Answers improved, the knowledge base has no size limit and latency is much lower.
- **Security and reliability:** Fixed critical issues in an AI-generated codebase: unauthorized admin access, sensitive-data exposure, Service Worker caching problems and inconsistent data from multiple sources of truth.
- **Refactor:** Led a large refactor: modularised the app, extracted reusable components and replaced hard-coded values with an admin-configurable system.
- **Monitoring:** Implemented monitoring and observability for faster root-cause analysis.

Mindsight — Full Stack Engineer

May 2024 – Feb 2026 · Remote

- **Multi-tenant HR platforms:** Delivered features, refactored legacy services and improved stability on platforms serving tens of thousands of monthly users.
- **Performance:** Refactored critical processing flows, cutting memory use and execution time by more than 50%.
- **Authentication and compliance:** Built SSO with external IdPs, OAuth2 between internal systems, and event journaling for compliance.
- **Job board:** Built a job-board application end to end (Django + React) in a team of two, owning architecture, data modelling, API design, testing and deployment.

Agile Content — Backend Software Engineer

Mar 2022 – May 2024 · São Paulo (Remote)

- **Subscription billing:** Built microservices (Node.js APIs and serverless functions) for the purchase, billing, renewal and cancellation flows of the largest telecom in Argentina.
- **AWS:** Designed AWS solutions (Lambda, SQS, ECS, CloudWatch, X-Ray, S3, Redshift, RDS) handling millions of events per month.
- **Infrastructure:** Worked with Docker, Terraform, the Serverless Framework and CI/CD.
- **Testing:** Led internal API-testing libraries in Robot Framework and reached 100% unit-test coverage.

IPQ Tecnologia — Full Stack Software Engineer

Feb 2021 – Mar 2022 · Salvador, Brazil

- **Public-security platform:** Developed and scaled a police-intelligence platform used by government agencies, which contributed to a reported crime reduction of more than 30%.
- **Kafka:** Built high-throughput Kafka integrations processing more than 2 million messages per day.
- **Core modules:** Built map layers, system integrations and confidential core modules under strict reliability requirements.

SENAI CIMATEC — Full Stack Developer Intern · Augmented Reality Intern

Feb 2018 – Aug 2019 · Salvador, Brazil

- **Industry and startup systems:** Built an API for vehicle-fleet management, a greenhouse sensor-monitoring system and traceability for wind-tower manufacturing.
- **Mixed reality:** Worked on a mixed-reality application for Microsoft HoloLens and created its i18n library.

EDUCATION

- **Postgraduate degree, Applied AI Engineering:** UNIPDS · 2026 – 2027 (in progress)
- **MBA, Software Engineering:** Universidade de São Paulo (USP) · 2023 – 2025. Thesis: serverless computing integrated with microservices.
- **B.Sc., Computer Science:** UNIFACS · 2022
- **Computer Engineering (transferred to Computer Science):** SENAI CIMATEC · 2017 – 2021. Coordinator of the competitive-programming group.
- **General English (C1):** Griffith College Cork, Ireland · 2019 – 2020

CERTIFICATIONS

- AI Agent Developer · Vanderbilt University (Coursera), 2026
- Generative AI Software Engineering · Vanderbilt University (Coursera), 2026
- HSK (Chinese Proficiency Test), 2022

SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL, C#, C++
- **Backend:** Node.js, NestJS, Express, Fastify, Django, REST, GraphQL, WebSockets, OpenAPI, Prisma
- **Frontend:** React, Vite, Tailwind, React Native, Electron
- **Data and messaging:** PostgreSQL (tuning, PgBouncer), Redis, MongoDB, DynamoDB, Kafka, RabbitMQ, BullMQ, SQS
- **Cloud and platform:** Kubernetes (k3s), KEDA, Kustomize, Docker, Ansible, Terraform, AWS (Lambda, ECS, SQS, S3, RDS), Jenkins, GitHub Actions
- **Observability and SRE:** OpenTelemetry, Prometheus, Thanos, Grafana, Grafana Faro, Loki, CloudWatch, SLOs and burn-rate alerting, runbooks, post-mortems, load testing
- **Security:** secrets management, Gitleaks, Semgrep, SSO/OAuth2, least-privilege access, audit logging, encryption of sensitive data
- **Testing:** Jest, Vitest, Mocha, Chai, Robot Framework, end-to-end and accessibility testing
- **AI:** LLM applications, RAG, vector databases, AI agents, LangChain, Claude Code, prompt engineering
- **Architecture:** modular monolith, microservices, event-driven systems, serverless, DDD, ADRs

LANGUAGES

Portuguese (native) · English (fluent, C1) · Spanish (fluent) · Mandarin Chinese (elementary, HSK) · Japanese (learning)
